

Mentora - AI Tutor for Personalized Learning



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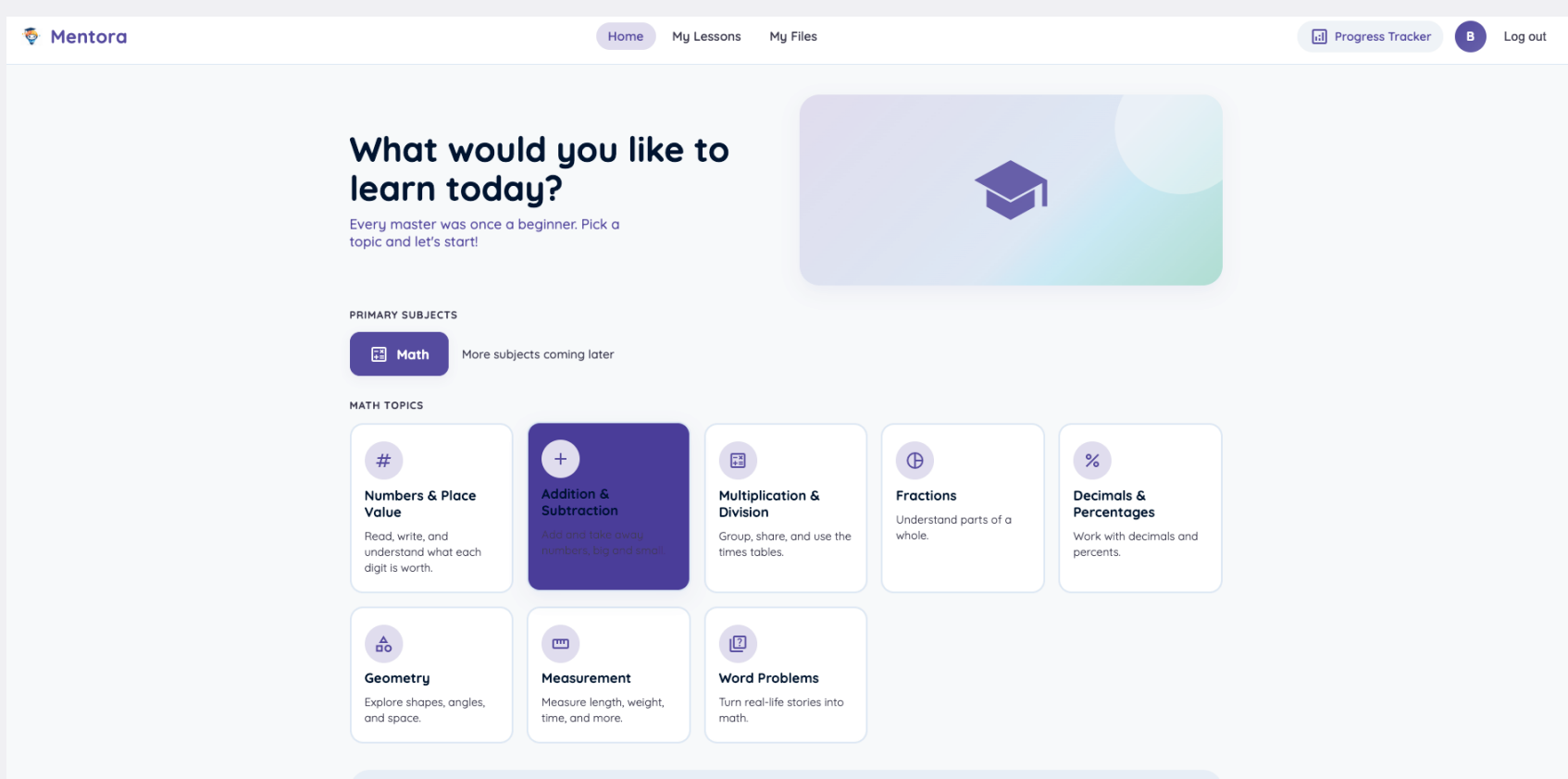
The Problem:

- 👥 One-size-fits-all learning
- 💰 Private tutor is expensive
- 📖 Online learning is passive
- 💬 No real-time feedback

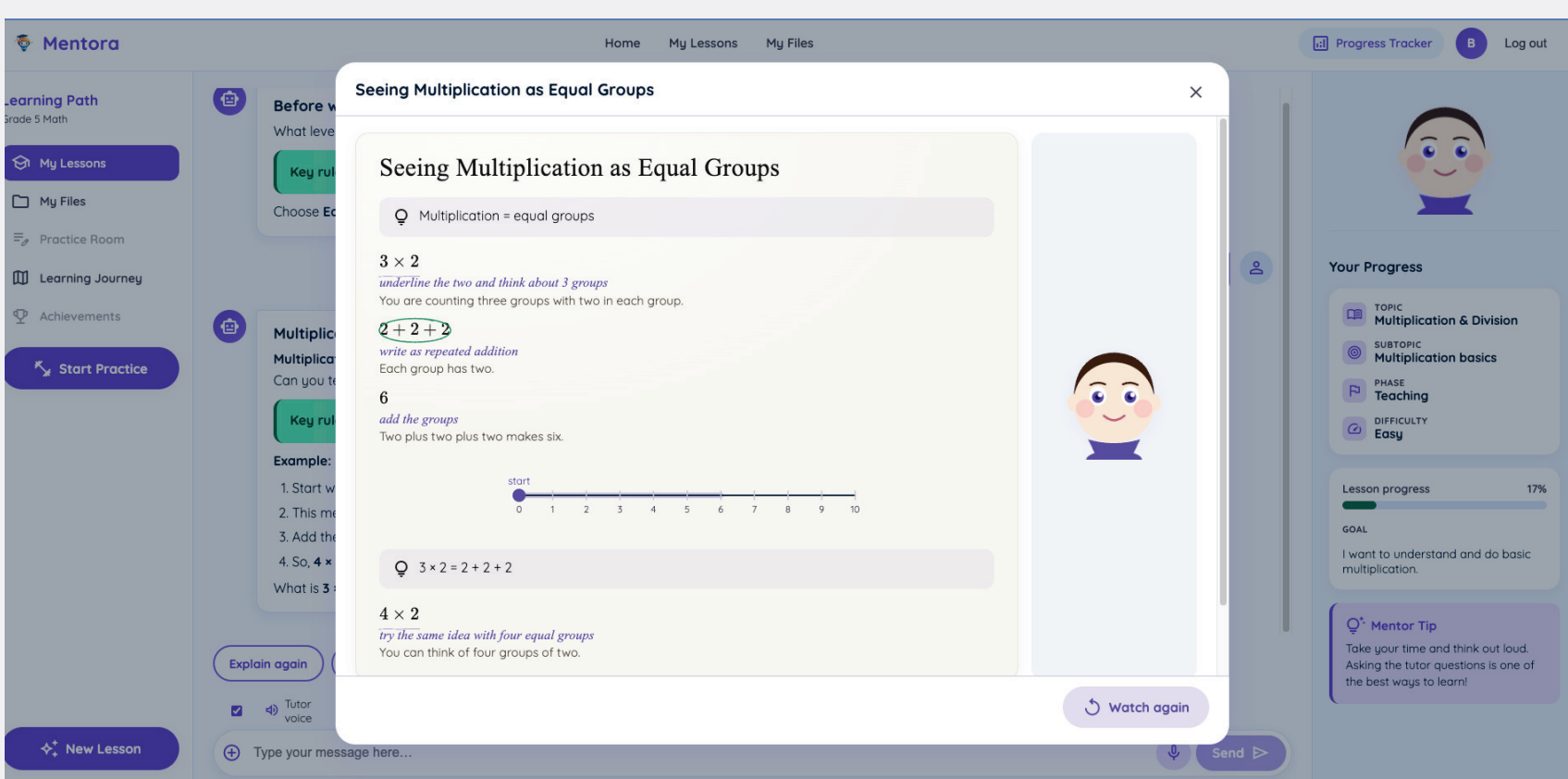
Meet Mentora

An AI-powered conversational tutor that explains concepts, ask questions, and adapts in real time.

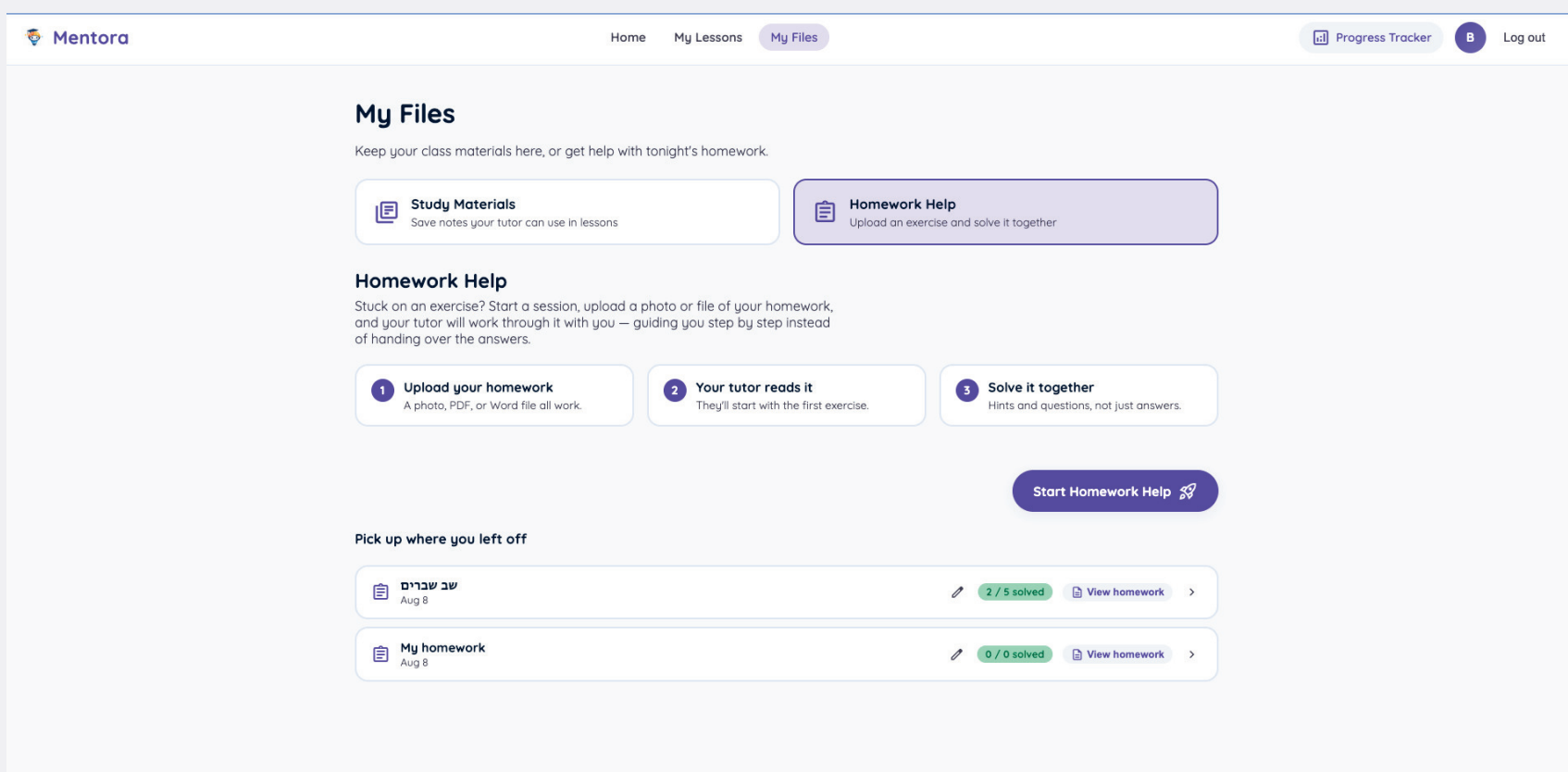
Subject & Topic Selection



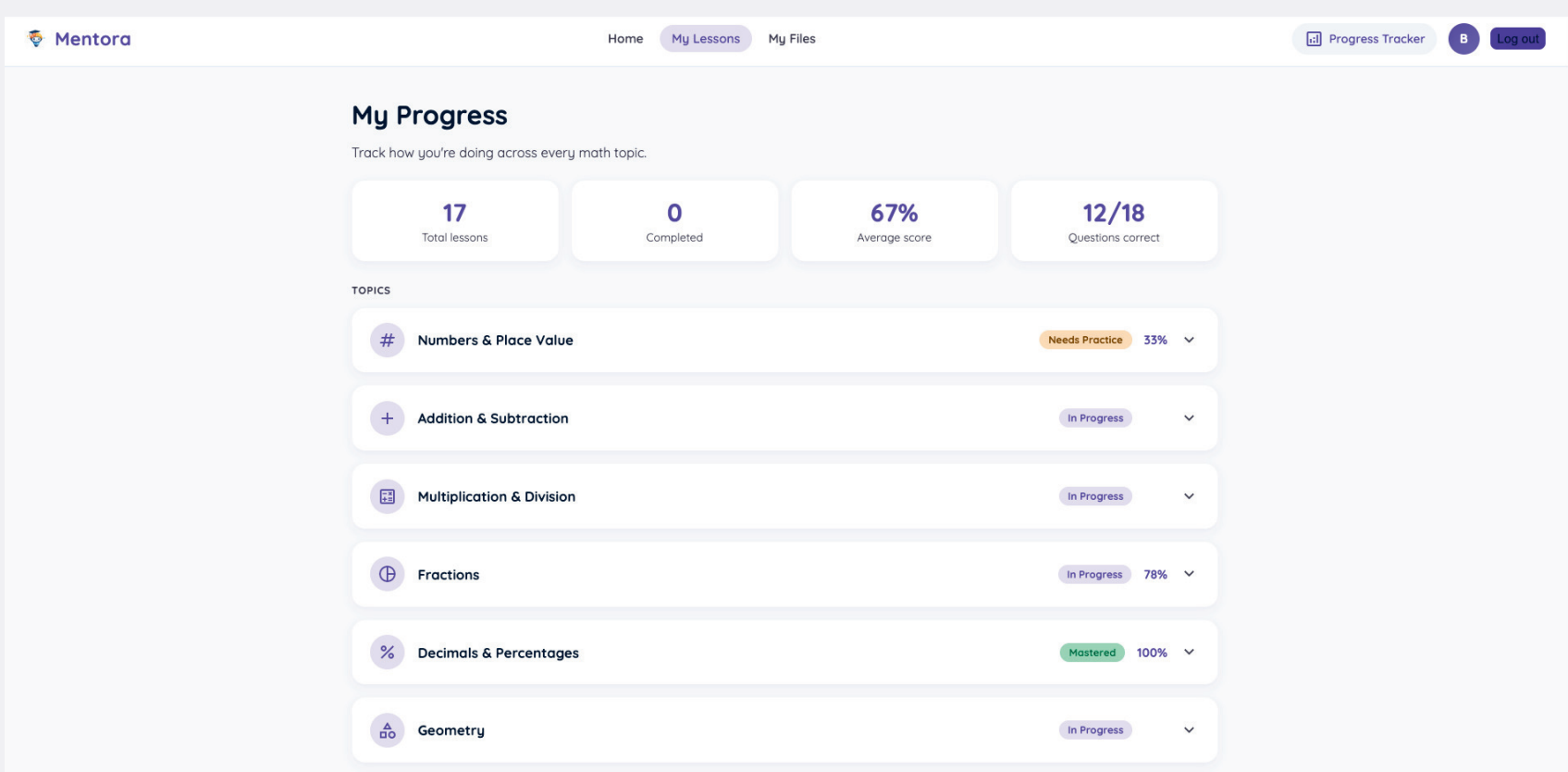
Conversational Tutor



Homework & Personalized teaching



Progress Tracking

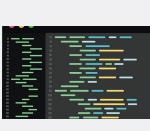


System Architecture



Frontend

React 18, TypeScript,
Vite, React Router



Backend

Python, FastAPI



AI Integration

OpenAI: Personalized tutoring, content generation & feedback
Multimodal AI: Vision + Voice interaction



Core Modules

Tutoring, Practice, Homework,
Progress, Study Materials,
Voice

Data & Storage

SQL Database

Users, sessions,
progress & assessments

File Storage

Study materials &
homework

API Layer

REST API

Frontend ↔ Backend

JWT

Authentication

AI & Database

Integration